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RESEARCH ARTICLE

A Lightweight Heterogeneous Neural Network Architecture for Multiscale Feature Extraction in Bearing Fault Diagnosis

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ABSTRACT Bearings are critical components in rotating machinery, and efficient fault detection is essential for maintaining motor reliability and operational integrity. This study proposes a novel neural network architecture for efficient bearing fault diagnosis that simultaneously captures local and global feature representations from log-Mel spectrograms derived from wideband acoustic emission signals. The framework combines a multiscale convolutional neural network with a Mamba-based selective state space model to improve the discrimination of complex fault patterns while preserving low computational complexity. To support context-aware learning, an adaptive feature fusion strategy is adopted to dynamically integrate features from preceding layers. The proposed framework is further implemented on an embedded platform to evaluate its suitability for edge computing. Experimental results demonstrate that the method achieves superior accuracy and efficiency across multiple fault types and rotational speeds, outperforming existing classification approaches. These results support the development of fully automated bearing condition monitoring systems and underscore the potential of in-sensor computing.

INDEX TERMS Acoustic emission, convolutional neural networks, condition monitoring, deep learning, digital signal processing, feature fusion, fault diagnosis, multiscale extraction, rotating machines, state space model, spectrogram.

I. INTRODUCTION

Bearings are pivotal components in mechanical systems, responsible for supporting loads and facilitating smooth rotational motion with minimal friction [1]. However, they are inherently susceptible to faults that can trigger catastrophic failures, unplanned production stoppages, and hazardous operating conditions [2]. Consequently, the demand for accurate bearing fault detection has intensified to mitigate operational risks and prevent severe failures. Increasing industrial complexity and system scale have exposed the limitations of manual inspection methods, which are typically labor-intensive, subjective, and inconsistent. In response, automated diagnostic approaches that integrate sensor-based data acquisition with advanced signal processing and machine learning techniques have gained widespread attention for their improved reliability and efficiency [3].

In the field of structural health monitoring, acoustic emission (AE) signals have emerged as a powerful diagnostic modality due to their nonintrusive nature and ability to reveal characteristic fault signatures, particularly in environments where vibration signals are masked by noise or interference. AE signals are inherently nonstationary and consist of high-frequency transient bursts associated with microstructural damage events. Liu et al. demonstrated that AE-based methods outperform vibration-based measurements under identical fault conditions at elevated rotational speeds [4]. Similarly, Cornel et al. reported that AE systems can detect pitting damage approximately 4% earlier than vibration-based approaches [5]. Despite these advantages, the complex

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and dynamic spectral characteristics of AE signals require advanced analytical methods capable of capturing transient behavior and evolving frequency content.

Classical spectral analysis is ill-suited for nonstationary signal processing because it neglects temporal variations. Although the Fourier transform is widely used to map signals from the time domain to the frequency domain, it exhibits fundamental limitations when applied to complex AE signals that are critical for accurate fault identification. To address these shortcomings, time–frequency analysis techniques have been developed to provide joint time and frequency resolution. Among these, the short-time Fourier transform (STFT) applies finite-length windows to approximate local stationarity, generating spectrograms that describe the temporal evolution of frequency components [6]. For example, Liu et al. employed STFT to produce two-dimensional (2-D) representations from raw signals for subsequent analysis [7]. Building on this approach, wavelet transforms offer multi-resolution analysis, enabling simultaneous detection of short-duration, high-frequency events and long-duration, low-frequency trends. Similarly, empirical mode decomposition (EMD) and the Hilbert–Huang transform (HHT) adaptively decompose signals into intrinsic mode functions, capturing nonlinear and nonstationary characteristics without predefined basis functions. Li et al. proposed a fast and adaptive EMD framework that efficiently isolates salient fault-related features while maintaining low computational cost [8]. Further developments, such as variational mode decomposition (VMD), improve decomposition accuracy and noise robustness, enhancing reliability in practical diagnostic settings [9]. Jian et al. applied VMD to process motor-bearing fault signals and optimized its parameters using the secretary bird optimization algorithm, resulting in improved robustness and decomposition precision [10].

Over the past decade, deep learning has driven significant advances in machinery fault diagnosis. Convolutional neural networks (CNNs), in particular, have proven effective for high-accuracy bearing fault classification by autonomously extracting discriminative features from visual and time–frequency representations of sensor data. Rajabioun et al. trained a CNN on 2-D signal images from multiple sources and achieved a strong diagnostic performance [11]. Hoang and Kang proposed a CNN-based framework with information fusion for motor-current signals and validated its effectiveness using empirical testing of real bearing fault data [12]. Wang et al. introduced a damage dynamics CNN to enhance interpretability while maintaining high classification accuracy [13]. Nevertheless, conventional single-scale CNNs are limited by fixed receptive fields, restricting their ability to capture hierarchical and scale-varying fault characteristics, particularly under noisy conditions.

To overcome these limitations, multiscale CNN architectures have been increasingly adopted. By employing convolutional kernels of different sizes across parallel branches, multiscale CNNs can simultaneously extract fine-grained local features and broader contextual information. Atrous convolution is another common strategy that expands the receptive field without increasing parameter count, improving robustness and representation quality under severe noise contamination. Yin et al. proposed a multiscale graph convolutional network to capture latent relationships in vibration signals across scales [14]. Shi et al. developed an enhanced lightweight multiscale CNN for rolling-bearing fault diagnosis, demonstrating improved feature discrimination and strong anti-noise performance [15]. Integrating attention mechanisms with multiscale frameworks further enhances generalization by enabling dynamic emphasis on informative features while suppressing irrelevant responses. For example, Shen and Zhang combined a multiscale CNN with an attention mechanism to exploit inter-channel interactions and capture localized dependencies [16]. Li et al. proposed an adaptive multiscale fully convolutional network using large-kernel convolutions and stacked residual adaptive blocks, achieving superior diagnosis performance under noisy conditions [17]. Hu et al. incorporated multiple attention mechanisms into a multiscale CNN to improve learning of critical fault features [18], while Zhao et al. integrated a multiscale CNN with bidirectional long short-term memory and attention mechanisms to achieve strong fault diagnosis performance across operating conditions [19]. Sun et al. proposed WFFNet, a novel neural network architecture that integrates convolutional operations, squeeze-and-excitation (SE) attention mechanism, and a weighted feature fusion strategy to enhance the network's ability to capture complex patterns, thereby improving feature learning capacity and overall performance [20]. Xu et al. developed TFDDP, an advanced architecture that leverages an encoder designed with multiple attention mechanisms in conjunction with residual network structures, thereby offering a more robust solution even with limited labeled samples [21].

Although these approaches have achieved notable success in local feature extraction, they remain limited in modeling long-range dependencies and global relationships. This shortcoming has motivated the development of hybrid architectures that combine complementary modeling paradigms. In particular, frameworks that integrate CNNs for local feature extraction with transformer-based models for long-range dependency modeling have gained traction. Wang et al. combined CNNs with transformers to achieve accurate and robust bearing-fault diagnosis [22]. Chen et al. proposed a cross-space multiscale CNN-transformer framework, demonstrating its ability to capture both spatial and temporal features [23]. Fang et al. designed a multiscale time–frequency CNN–transformer with attention mechanisms for practical fault diagnosis applications [24]. Han et al. introduced MT-ConvFormer, a hybrid CNN-transformer framework for multitask fault diagnosis [25]. Zhang et al. combined an adaptive multiscale CNN with self-attention to address the limitations of single-scale representations in noisy

environments [26]. He et al. introduced TFDDCF, a hybrid deep learning framework that integrates enhanced CNNs with a transformer, ultimately boosting the model's ability to learn complex structures and deliver superior performance [27].

Advancing from representation learning on transformed fault signals to architectures that integrate discriminative power with contextual awareness represents a promising trajectory in fault diagnosis research. Li et al. proposed the wavelet capsule network under conditions of incomplete data, enhancing model transparency and interpretability in diagnosis decision-making [28]. Huang et al. introduced a deep adversarial capsule network to incorporate multidomain generalization into compound fault diagnosis, addressing the challenge of absent labeled data under new or extreme working conditions [29]. Wang et al. developed a dynamic collaborative adversarial domain adaptation network that improves fault diagnosis in rotating machinery when labeled data for new tasks is limited [30]. Wang et al. proposed an adaptive fused domain-cycling variational generative adversarial network to mitigate the limitations imposed by data scarcity [31]. Wu et al. introduced a conditional distribution-guided adversarial transfer learning network to resolve the complexities of multisource domains utilization in fault diagnosis [32]. Xu et al. proposed a time-frequency fully connected graph neural network, which leverages temporal-spectral correlations to improve the accuracy of multisource fault diagnosis [33]. Huang et al. developed an Exofeature-aware transformer to enable long-horizon state-of-health prediction for batteries, achieving superior performance compared to CNN and Crossformer approaches [34].

Despite these advances, most existing fault diagnosis methods rely on computationally intensive hardware and incur substantial resource consumption, limiting their practicality for real-world deployment, particularly on embedded platforms with strict power constraints. Data preprocessing pipelines often impose significant computational overhead, and transformer-based models, while powerful, remain resource-demanding and complex.

To address these challenges, we propose a lightweight heterogeneous neural network architecture that integrates a multiscale CNN with a Mamba-based selective state space model (M-SSSM) operating in parallel. By leveraging the complementary strengths of these components, the proposed framework achieves improved diagnostic accuracy while maintaining computational efficiency. The multiscale CNN captures discriminative local features across multiple scales, while the M-SSSM extracts global contextual information and long-range dependencies. The outputs from both branches are dynamically combined through an adaptive feature fusion strategy to balance local and global representations. In addition, AE signals are preprocessed using STFT and transformed into log-Mel spectrograms for dimensionality reduction before being fed into the network. The resulting framework is well-suited for deployment on embedded systems under practical operating conditions.

The main contributions of this work are as follows.

1) We demonstrate that the proposed method consistently achieves high classification accuracy and strong robustness under noisy operating conditions.
2) We employ an efficient time–frequency analysis pipeline that converts nonstationary AE signals into 2-D representations using STFT and log-Mel mapping, yielding perceptually meaningful features with substantial dimensionality reduction.
3) We design a multiscale CNN architecture that captures diverse short-range dependencies with lower memory usage and computational complexity than conventional backbone CNN models.
4) We propose a heterogeneous architecture that combines a multiscale CNN with a customized M-SSSM to jointly extract local and global features, achieving accuracy and efficiency relative to existing hybrid approaches while remaining suitable for resource-constrained devices.
5) We validate the proposed framework through implementation on an embedded system under real-world operating conditions.

The remainder of this paper is organized as follows. Section II outlines the theoretical foundation of the study. Section III describes the proposed method in detail. Section IV presents the experimental setup and evaluation results. Finally, Section V concludes the paper and discusses directions for future research.

II. PRELIMINARIES

This section introduces the essential background and key concepts that underpin this study, providing a clear and consistent foundation for understanding the proposed methodology and the experimental analyses presented later.

A. SHORT-TIME FOURIER TRANSFORM

Unlike the classical Fourier transform, which provides a global frequency representation of a signal, the STFT enables localized spectral analysis by applying the Fourier transform to short, overlapping segments of a nonstationary signal [35]. This property makes the STFT particularly suitable for applications such as speech processing, biomedical signal analysis, and audio classification. In practice, analog signals are typically sampled, yielding a discrete-time representation. Mathematically, the discrete STFT process is defined as:

$$r_a^{\text{raw}}[m] = r^{\text{raw}}[m + aH]\, w[m] \tag{1}$$

$$S[k, a] = \frac{1}{M} \sum_{m=1}^{K} r_a^{\text{raw}}[m]\, e^{-\frac{j2\pi mk}{K}} \tag{2}$$

where r^{raw} is the analysis signal, w is the sliding window function, $a \in \{0, 1, \ldots, N-1\}$ is the frame index, $N \in \mathbb{N}$ is the number of frames, $H \in \mathbb{N}$ denotes the hop size, $m \in \{1, 2, \ldots, M\}$ represents the discrete time positions at which the sliding window is applied, $M \in \mathbb{N}$ is the window length, $k \in \{1, 2, \ldots, K\}$ is the frequency-bin index, and

$K \in \mathbb{N}$ is the total number of discrete Fourier transform (DFT) points. Each value $S[k, a]$ represents the complex amplitude at frequency bin k and time frame a, forming an individual element of the resulting STFT matrix.

B. CONVOLUTIONAL NEURAL NETWORK

CNNs are a class of feedforward neural networks that employ convolutional operations to automatically extract spatially localized features from input data [36]. This architectural paradigm has demonstrated strong performance in computer vision tasks, including image classification, object detection, and semantic segmentation [37]. Beyond vision, CNNs have also been successfully applied to natural language processing, speech recognition, and audio signal analysis [38], highlighting their versatility as a general-purpose feature learning framework.

Let $X^{(l-1)} \in \mathbb{R}^{U_{l-1} \times V_{l-1} \times G_{l-1}}$ be a 3-D input tensor to the l-th convolutional layer, where U_{l-1}, V_{l-1}, and G_{l-1} represent the height, width, and number of channels, respectively. The layer comprises a set of G_l learnable kernels $W^{(l,c)} \in \mathbb{R}^{K_{\mathrm{h}}^{(l)} \times K_{\mathrm{w}}^{(l)} \times G_{l-1}}$, where $K_{\mathrm{h}}^{(l)}$ and $K_{\mathrm{w}}^{(l)}$ denote the height and width of the kernel, and $c \in \{1, 2, \ldots, G_l\}$ indexes the individual filter. The pre-activation feature map $Z^{(l,c)}$ at spatial position (i, j) using stride $(s_{\mathrm{h}}^{(l)}, s_{\mathrm{w}}^{(l)})$ with corresponding bias $b^{(l,c)} \in \mathbb{R}$, is given by:

$$Z_{i,j}^{(l,c)} = \sum_{u=0}^{K_{\mathrm{h}}^{(l)}-1} \sum_{v=0}^{K_{\mathrm{w}}^{(l)}-1} \sum_{r=1}^{G_{l-1}} W_{u,v,r}^{(l,c)} X_{is_{\mathrm{h}}^{(l)}+u, js_{\mathrm{w}}^{(l)}+v, r}^{(l-1)} + b^{(l,c)} \quad (3)$$

To improve optimization and stabilize training, batch normalization is commonly applied to each convolutional channel. For a given channel c in layer l, batch normalization is defined as:

$$\hat{Z}_{i,j}^{(l,c,n)} = \gamma^{(l,c)} \frac{Z_{i,j}^{(l,c,n)} - \mu^{(l,c)}}{\sqrt{(\sigma^{(l,c)})^2 + \varepsilon}} + \beta^{(l,c)} \quad (4)$$

where n indexes the mini-batch examples, $\gamma^{(l,c)}$ and $\beta^{(l,c)}$ are learnable parameters, $\varepsilon > 0$ is a small constant for numerical stability, $\mu^{(l,c)}$ and $(\sigma^{(l,c)})^2$ are the batch mean and variance, respectively:

$$\mu^{(l,c)} = \frac{1}{\left|\left\{Z_{i,j}^{(l,c,n)}\right\}\right|} \sum_{(i,j,n)} Z_{i,j}^{(l,c,n)} \quad (5)$$

$$(\sigma^{(l,c)})^2 = \frac{1}{\left|\left\{Z_{i,j}^{(l,c,n)}\right\}\right|} \sum_{(i,j,n)} \left(Z_{i,j}^{(l,c,n)} - \mu^{(l,c)}\right)^2 \quad (6)$$

An activation function $\Phi(\cdot)$ is then applied to introduce nonlinearity into the network:

$$Y_{i,j}^{(l,c,n)} = \Phi\left(\hat{Z}_{i,j}^{(l,c,n)}\right) \quad (7)$$

In layer l, pooling operations $\Psi(\cdot)$ are applied independently for each n and c to reduce the spatial resolution of feature maps while preserving the most salient information. Assuming a pooling window of size $P_{\mathrm{h}} \times P_{\mathrm{w}}$ and stride $(t_{\mathrm{h}}, t_{\mathrm{w}})$, the pooling region at position (u, v) corresponds to the set:

$$R_{u,v}^{(l)} = \left\{ Y_{i,j}^{(l,c,n)} | i \in \{ut_{\mathrm{h}}, \ldots, ut_{\mathrm{h}} + P_{\mathrm{h}} - 1\}, \; j \in \{vt_{\mathrm{w}}, \ldots, vt_{\mathrm{w}} + P_{\mathrm{w}} - 1\} \right\} \quad (8)$$

The pooled output is then computed as:

$$P_{u,v}^{(l,c,n)} = \Psi\left(R_{u,v}^{(l)}\right) \quad (9)$$

After several convolutional layers, the resulting feature maps are flattened into a 1-D vector and passed to a fully connected (FC) layer for higher-level representation learning:

$$h_{\mathrm{out}}^{(n)} = \mathrm{vec}\left(P^{(l,n)}\right) \quad (10)$$

$$o^{(n)} = \Phi\left(h_{\mathrm{out}}^{(n)} W_{\mathrm{fc}}^{\mathrm{T}} + b_{\mathrm{fc}}\right) \quad (11)$$

where W_{fc}, b_{fc}, and $o^{(n)}$ denote the weight, bias, and output of the FC layer, respectively.

C. STATE SPACE MODEL

State space models (SSMs) provide a general mathematical framework for describing the evolution of dynamic systems through latent states and observable outputs [39]. In this formulation, a system is characterized by a set of state variables that encapsulate the information required to predict future behavior. Classically, an SSM is defined by two equations: a process equation that governs state evolution and an observation equation that maps latent states to measured outputs:

$$\dot{h}(t) = Ah(t) + Bx(t) \quad (12)$$

$$y(t) = Ch(t) \quad (13)$$

where $x(t)$ and $y(t)$ represent the system input and output, respectively, $h(t)$ is the state vector, A describes the autonomous state dynamics, B maps the input to the state, and C relates the state to the observation.

When the input is a discrete sequence rather than a continuous time signal $x(t)$, the SSM must be discretized using a step size Δ that specifies the input resolution:

$$h_q = \bar{A}h_{q-1} + \bar{B}x_q \quad (14)$$

$$y_q = Ch_q \quad (15)$$

$$\bar{A} = \exp(\Delta A) \quad (16)$$

$$\bar{B} = (\Delta A)^{-1} (\exp(\Delta A) - I) \cdot \Delta B \quad (17)$$

where $\bar{A}$ and $\bar{B}$ are discrete-time approximations of matrix A and B, respectively, using the zero-order hold (ZOH) discretization rule [40]. Eqs. (14) and (15) define the SSM in a recurrent form.

Modern machine learning frameworks often reinterpret SSMs as sequence-to-sequence convolutional operators. Given the sequential nature of the data, the SSM can be computed via a global 1-D convolution:

$$\bar{K} = \left(C\bar{B}, C\bar{A}\bar{B}, \ldots, C\bar{A}^q\bar{B}, \ldots\right) \quad (18)$$

$$y = x * \bar{K} \quad (19)$$

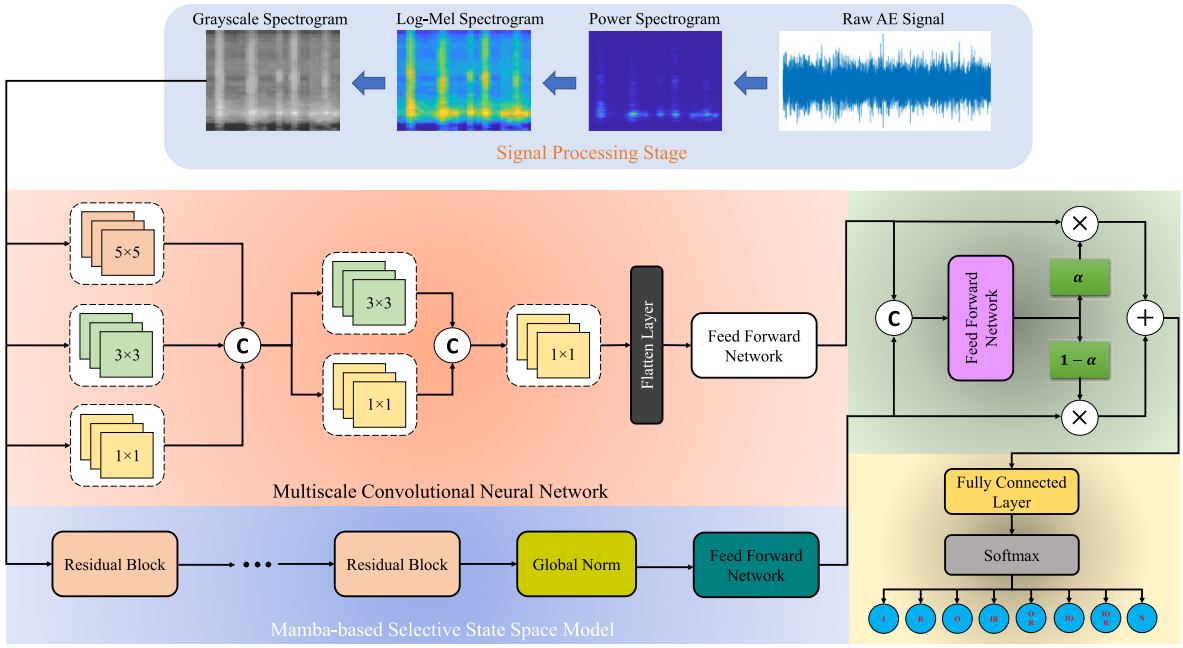


FIGURE 1. Proposed bearing fault diagnosis framework: Two main stages include signal processing and fault classification through a heterogeneous network. The raw AE signals are transformed into 2-D representations using time–frequency analysis and subsequently fed into a hybrid neural network to accurately and efficiently classify diverse fault types.

where $\bar{K}$ denotes a structured convolutional kernel.

During training, the model typically operates in convolutional mode, which allows for efficient parallel computation when the entire input sequence is available. During inference, however, the model switches to a recurrent mode, enabling efficient autoregressive processing by consuming one input element at a time.

III. PROPOSED METHOD

In this section, we present a comprehensive examination of the proposed methodology for bearing fault diagnosis, as illustrated in Fig. 1. The framework is organized into two primary stages, including signal processing and fault classification using a heterogeneous neural network.

Stage 1: Raw signals are collected using an AE sensor mounted on the bearing housing. These signals are transformed into 2-D spectrograms for feature extraction in the next stage.

Stage 2: The generated spectrograms are used as inputs to a heterogeneous neural network architecture designed to extract both local and global discriminative features. The architecture integrates a multiscale CNN with a customized M-SSSM. Outputs from the two branches are dynamically fused to balance the contributions of local and global representations. The fused features are then forwarded to the classification module, which serves as the decision-making component of the proposed framework.

A. TIME-FREQUENCY ANALYSIS

After acquisition via the AE sensor, the signals are segmented and transformed into 2-D representations using Eqs. (1) and (2), with a Blackman window length of 1024, a hop size of 256, and 4096 DFT points. The resulting STFT matrices are converted into power spectrograms as follows:

$$\tilde{S}[k,a] = \left(\frac{2}{\sum_{m=1}^{M} \frac{w[m]}{M}} |S[k,a]| \right)^2 \tag{20}$$

To reduce dimensionality, the spectrograms are mapped to the Mel scale using Mel filters. The Mel scale converts linear frequency f to a perceptual frequency n_{mel} as follows:

$$n_{\text{mel}}(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right) \tag{21}$$

with the inverse mapping given by:

$$f(n_{\text{mel}}) = 700(10^{n_{\text{mel}}/2595} - 1) \tag{22}$$

Each Mel filter $F_{n_{\text{mel}}}[k]$ is defined as a triangular weighting function over the frequency bins:

$$F_{n_{\text{mel}}}[k] = \begin{cases} \dfrac{k - k_{n_{\text{mel}}-1}}{k_{n_{\text{mel}}} - k_{n_{\text{mel}}-1}}, & k_{n_{\text{mel}}-1} \le k \le k_{n_{\text{mel}}} \\ \dfrac{k_{n_{\text{mel}}+1} - k}{k_{n_{\text{mel}}+1} - k_{n_{\text{mel}}}}, & k_{n_{\text{mel}}} \le k \le k_{n_{\text{mel}}+1} \\ 0, & \text{otherwise} \end{cases} \tag{23}$$

The Mel spectrograms are generated by applying 128 Mel filters to the power spectrograms:

$$\tilde{X}[n_{\text{mel}}, a] = \sum_{k=1}^{K} F_{n_{\text{mel}}}[k] \tilde{S}[k,a] \tag{24}$$

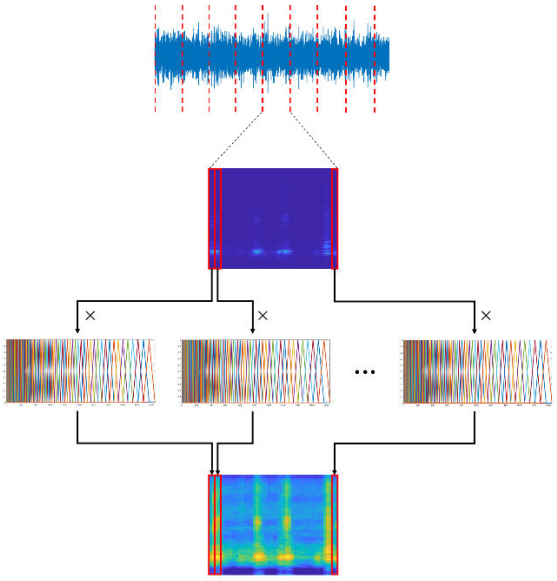


FIGURE 2. Overall signal processing pipeline: the acquired signal is first segmented into smaller frames, after which each frame is transformed into a power spectrogram using the STFT. The final spectrogram is generated by applying Mel filters and logarithmic compression.

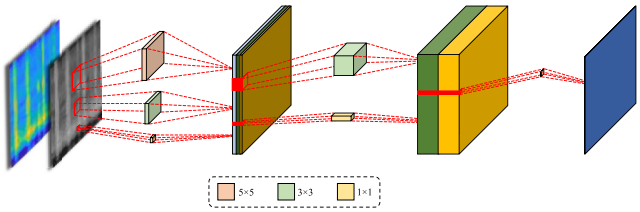


FIGURE 3. The multiscale feature extraction process integrates information captured at different receptive fields. The final output retains only the most salient and discriminative feature.

TABLE 1. Configuration of the proposed multiscale CNN.

Layer	Operation	Kernel size/ Filter number	Stride	Padding	Output shape
1	Conv2d, BN, ReLU	5/20	1	2	20×32×32
	Conv2d, BN, ReLU	3/20	1	1	20×32×32
	Conv2d, BN, ReLU	1/8	1	0	8×32×32
2	Conv2d, BN, ReLU	3/32	1	1	32×32×32
	Conv2d, BN, ReLU	1/32	1	0	32×32×32
3	Conv2d, BN	1/1	1	0	1×32×32
4	Flatten	-/-	-	-	1×1024
5	FC (256), ReLU, Dropout (0.5)	-/-	-	-	1×256
6	FC (256), ReLU	-/-	-	-	1×256

CONV2D IS A 2-D CONVOLUTIONAL OPERATION. THIS TABLE ASSUMES THAT THE INPUT IMAGE SIZE IS 32×32.

To improve numerical stability for the subsequent model processing, logarithmic compression is applied:

$$X[n_{\text{mel}}, a] = 20 \log_{10} \left(\tilde{X}[n_{\text{mel}}, a] + 10^{-6} \right) \tag{25}$$

The final log-Mel spectrograms are then converted into single-channel grayscale representations and resized to 32 × 32 for efficient downstream processing. This resizing simplifies data preparation while preserving essential information. The overall data preprocessing stage is illustrated in Fig. 2.

B. MULTISCALE CONVOLUTIONAL NEURAL NETWORK

Our multiscale CNN architecture is designed to capture local features from generated spectrograms at multiple spatial resolutions. By employing convolutional kernels of varying sizes, the network simultaneously extracts fine-grained details and broader contextual patterns inherent in the spectrogram, as shown in Fig. 3.

In the first convolutional layer, three kernel sizes (5 × 5, 3×3, and 1×1) are used in parallel. The 5×5 kernels provide a larger receptive field, the 3 × 3 kernels focus on intermediate-scale features, and the 1 × 1 kernels perform pointwise operations across pixel intensities. Each convolutional layer is followed by batch normalization (BN), as defined in Eq. (4), and a rectified linear unit (ReLU) activation function:

$$Y_{i,j}^{(l,c,n)} = \Phi\left(\hat{Z}_{i,j}^{(l,c,n)}\right) = \frac{\hat{Z}_{i,j}^{(l,c,n)} + \left|\hat{Z}_{i,j}^{(l,c,n)}\right|}{2} \tag{26}$$

where $\Phi(\cdot)$ denotes the activation function, and $\hat{Z}_{i,j}^{(l,c,n)}$ represents the pre-activation feature map, both defined in Section II.

Feature maps generated by the first layer are concatenated along the channel dimension, merging outputs from different kernel scales into a unified representation. As a result, the output dimensionality increases, with the total number of channels equal to the sum of channels produced by each kernel branch.

The second convolutional layer continues hierarchical feature extraction by refining and extending representations from the previous layer. It applies two kernel sizes (3 × 3 and 1 × 1) in parallel and concatenates their outputs along the channel dimension to enrich the feature space.

Following concatenation, a pointwise operation is applied to the aggregated feature maps to reduce dimensionality. This operation compresses the channel dimension, emphasizing salient features while suppressing redundancy, thereby maintaining computational efficiency without sacrificing discriminative capability.

The reduced feature map is then flattened into a 1-D vector suitable for dense processing. The flattened vector is passed through two FC layers, each with 256 neurons. To improve generalization and mitigate overfitting, a dropout layer is inserted between the FC layers, randomly deactivating a subset of neurons during training. The detailed architectural parameters of the proposed multiscale CNN are summarized in Table 1.

C. MAMBA-BASED SELECTIVE STATE SPACE MODEL

Recent developments have shown that SSMs provided an efficient alternative to attention mechanisms for

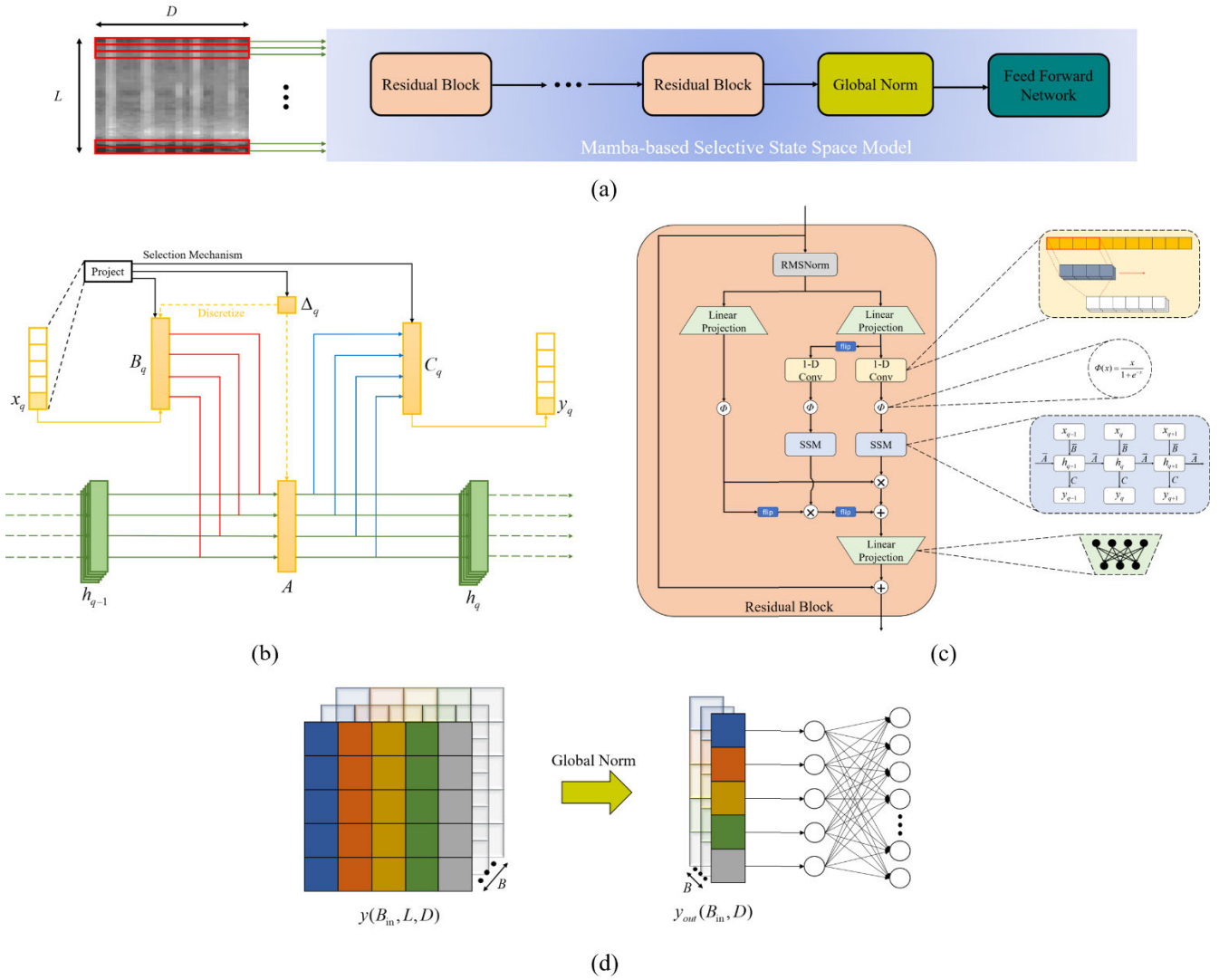


FIGURE 4. M-SSSM process. (a) Each row of an input image is interpreted as a token, where *D* denotes the feature dimension and *L* corresponds to the sequence length. The M-SSSM architecture is composed of 24 stacked residual blocks and a multilayer perceptron. (b) The selective SSM has a selection mechanism in which the selective scan algorithm dynamically filters irrelevant information by adapting transition pathways through matrices *B* and *C*. (c) Each residual block contains core operations: Root mean square layer normalization (RMSNorm), 1-D convolutional kernels, sigmoid linear unit activation functions, selective SSM blocks, and linear projection operations. (d) The Global Norm computes the mean over the sequence axis *L*, producing one *D*-dimensional vector per sample, where B_{in} represents the total number of samples (batch size). The resulting vectors are subsequently passed into a feedforward layer, which refines condensed global features and enhances their discriminative capacity.

sequence-modeling tasks [40]. By incorporating selective gating mechanisms (Fig. 4b), selective SSMs dynamically regulate state transitions according to input relevance, thereby enhancing contextual awareness [41]. In computer vision, M-SSSMs have demonstrated remarkable performance, highlighting their suitability for global feature extraction and complex visual recognition tasks [42], [43], [44].

In this study, we integrate a multiscale CNN with a customized M-SSSM to capture both local and global fault feature relationships. As illustrated in Fig. 4b, the selective scan mechanism in M-SSSM dynamically generates input-dependent parameters from the AE features, enabling adaptive state updating and output projection at each timestep. This design allows the model to selectively retain informative fault transients while suppressing lengthy background intervals. This approach provides a more suitable inductive bias than fixed state models or dense self-attention. Consequently, the M-SSSM can simultaneously achieve efficient long-sequence modeling and enhanced sensitivity to weak transient fault signatures. In the M-SSSM branch, the spectrogram is not directly flattened into a 1-D sequence of pixels. Instead, each row of the spectrogram is treated as a token, where each token represents the temporal evolution of a specific frequency band. The sequence is then constructed along the frequency dimension, enabling the model to capture dependencies across different frequency components.

Since each token retains its internal temporal structure, this representation preserves essential time-frequency information while allowing efficient sequence modeling. This design is particularly suitable for AE signals, where fault-induced energy is distributed across multiple frequency bands. Within the M-SSSM architecture, 24 residual blocks are stacked sequentially for a favorable trade-off between performance and energy consumption (Fig. 4a). Each block receives sequence tokens as inputs, refines the representations, and forwards the enhanced features to the subsequent block.

Given a batch of single-channel input images $X_{\text{in}} \in \mathbb{R}^{B_{\text{in}}\times 1\times U_{\text{in}}\times V_{\text{in}}}$, where B_{in} is the number of batches, U_{in} and V_{in} represent the height and width, respectively, each image row is treated as a token in a 1-D sequence. The spatial grid is reshaped into a sequence of row vectors, yielding $x \in \mathbb{R}^{B_{\text{in}}\times L\times D}$, where L (equals to U_{in}) is the sequence length, and D (equals to V_{in}) is the token dimension. Within each residual block, root mean square layer normalization (RMSNorm) is applied across the feature dimension for each token. For a token vector $x^{(n,i_s)} \in \mathbb{R}^D$ at batch index n and sequence index i_s, the RMSNorm is defined as:

$$\widehat{x}^{(n,i_s)} = \gamma \frac{x^{(n,i_s)}}{\sqrt{\frac{1}{D}\sum_{d=1}^{D}\left(x_d^{(n,i_s)}\right)^2 + \varepsilon}} \tag{27}$$

where $\gamma \in \mathbb{R}^D$ is a learnable scale parameter, and $\varepsilon = 10^{-5}$ is a small constant introduced for numerical stability. Stacking all tokens yields:

$$\widehat{x} = \text{RMSNorm}(x) \in \mathbb{R}^{B_{\text{in}}\times L\times D} \tag{28}$$

The normalized representation $\widehat{x}$ is then projected using an expansion factor of $E = 2$ into two parallel embeddings $z, \tilde{x} \in \mathbb{R}^{B_{\text{in}}\times L\times ED}$, with z is a gating tensor, and $\tilde{x}$ is forwarded to the sequence processing branch.

To capture bidirectional contextual information along the row sequence, two parallel branches operate on $\tilde{x}$. The first branch processes the sequence in its original order using 1-D convolutional kernels of size four, followed by a sigmoid linear unit (SiLU) nonlinearity:

$$x_{\text{out}}^{(1)} = \text{SiLU}(\text{Conv1D}(\tilde{x})) \in \mathbb{R}^{B_{\text{in}}\times L\times ED} \tag{29}$$

Here, $x_{\text{out}}^{(1)}$ is the post-activation output that is passed through the SSM, where a selective scan algorithm dynamically adapts state transitions to preserve relevant information:

$$\widehat{y}^{(1)} = \text{SSM}\left(x_{\text{out}}^{(1)}\right) \in \mathbb{R}^{B_{\text{in}}\times L\times ED} \tag{30}$$

where $\widehat{y}^{(1)}$ denotes the SSM output. The output of the reverse-direction branch is then computed as:

$$\tilde{x}^{\text{rev}} = \text{flip}(\tilde{x}, \text{dim} = L) \in \mathbb{R}^{B_{\text{in}}\times L\times ED} \tag{31}$$

$$z^{\text{rev}} = \text{flip}(z, \text{dim} = L) \in \mathbb{R}^{B_{\text{in}}\times L\times ED} \tag{32}$$

where $\tilde{x}^{\text{rev}}$, z^{rev} are the flipped input to the second branch and the flipped gating tensor, respectively. Similarly, the output of the second branch is obtained as:

$$\widehat{y}^{(2)} = \text{SSM}\left(\text{SiLU}\left(\text{Conv1D}\left(\tilde{x}^{\text{rev}}\right)\right)\right) \in \mathbb{R}^{B_{\text{in}}\times L\times ED} \tag{33}$$

The outputs of the forward and reverse branches are gated using z and z^{rev}, respectively, and aggregated as follows:

$$\widehat{y} = \frac{\widehat{y}^{(1)} \odot \text{SiLU}(z) + \text{flip}\left(\widehat{y}^{(2)} \odot \text{SiLU}\left(z^{\text{rev}}\right), \text{dim} = L\right)}{2} \tag{34}$$

The aggregated output $\widehat{y} \in \mathbb{R}^{B_{\text{in}}\times L\times ED}$ is subsequently linearly projected back to the original feature dimension $\tilde{y} \in \mathbb{R}^{B_{\text{in}}\times L\times D}$ and combined with the residual input $x \in \mathbb{R}^{B_{\text{in}}\times L\times D}$ to form the block output $y \in \mathbb{R}^{B_{\text{in}}\times L\times D}$, which is propagated to the next residual block. The core operations of each residual block are illustrated in Fig. 4c.

After processing through all residual blocks, global normalization is applied along the sequence dimension to produce the final sequence output $y_{\text{out}} \in \mathbb{R}^{B_{\text{in}}\times D}$:

$$y_{\text{out}}^{(n,d)} = \frac{1}{L}\sum_{i_s=1}^{L} y_{i_s}^{(n,d)} \tag{35}$$

As shown in Fig. 4d, the resulting tensor is passed through a linear layer with 256 units to generate feature representations suitable for the subsequent fusion stage:

$$\widehat{v} = \text{norm}\left(y_{\text{out}} W_0^{\text{T}} + b_0\right) \in \mathbb{R}^{B_{\text{in}}\times D_{\text{out}}} \tag{36}$$

where $W_0 \in \mathbb{R}^{D_{\text{out}}\times D}$, $b_0 \in \mathbb{R}^{D_{\text{out}}}$ are learnable parameters of the linear layer, respectively, and norm $(\cdot)$ is the Euclidean normalization.

D. ADAPTIVE FEATURE FUSION

Static fusion strategies fail to account for the varying importance of features extracted by different branches. To address this limitation, adaptive feature fusion introduces a learnable weighting mechanism that selectively integrates local and global representations into a unified feature space, dynamically balancing fine-grained sensitivity with global contextual awareness.

Let $\widehat{o} \in \mathbb{R}^{B_{\text{in}}\times D_{\text{out}}}$ be the normalized local features from the multiscale CNN branch, and $\widehat{v} \in \mathbb{R}^{B_{\text{in}}\times D_{\text{out}}}$ be the normalized global features extracted from the M-SSSM branch. An adaptive fusion gate is:

$$\alpha = \psi\left(\varphi\left(\left[\widehat{o} \parallel \widehat{v}\right] W_1^{\text{T}} + b_1\right) W_2^{\text{T}} + b_2\right) \in (0,1)^{B_{\text{in}}\times D_{\text{out}}} \tag{37}$$

where $\varphi(\cdot)$ is the ReLU function, $\psi(\cdot)$ is the sigmoid function, $\left[o \parallel \widehat{v}\right]$ denotes feature concatenation, $W_1 \in \mathbb{R}^{Q\times 2D_{\text{out}}}$, $W_2 \in \mathbb{R}^{D_{\text{out}}\times Q}$, $b_1 \in \mathbb{R}^Q$, $b_2 \in \mathbb{R}^{D_{\text{out}}}$ are learnable parameters, $D_{\text{out}} = 256$ is the output vector dimension to match the output dimension of the multiscale CNN branch, and $Q = 128$ is the hidden layer dimension.

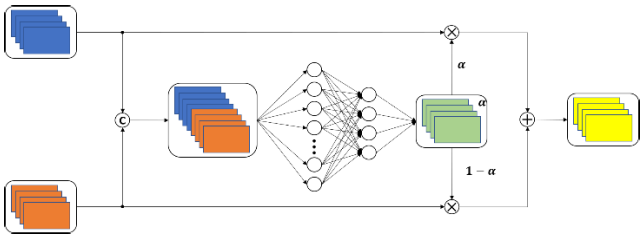


FIGURE 5. Adaptive feature fusion. Features from different branches are concatenated and then injected into a learnable network to produce gating vectors. These gates learn to balance outputs from branches, thereby improving the fusion method.

Subsequently, the two feature streams are combined through the gating vector:

$$\widehat{u} = \alpha \odot \widehat{o} + (1 - \alpha) \odot \widehat{v} \tag{38}$$

where $\widehat{u} \in \mathbb{R}^{B_{\text{in}} \times D_{\text{out}}}$ is the fused representation. Fig. 5 illustrates the adaptive fusion method used in this study.

E. CLASSIFIER AND LOSS FUNCTION

The fused output $\widehat{u}$ is subsequently processed through an FC transformation and a classifier to produce a probability distribution over fault classes:

$$\widehat{p} = \xi\left(\varphi\left(\widehat{u}W_{\text{fcn}}^{\text{T}} + b_{\text{fcn}}\right)W_{\text{cls}}^{\text{T}} + b_{\text{cls}}\right) \in (0, 1)^{B_{\text{in}} \times D_{\text{cls}}} \tag{39}$$

where $\xi(\cdot)$ is the softmax function, $W_{\text{fcn}} \in \mathbb{R}^{D_{\text{fcn}} \times D_{\text{out}}}$ and $b_{\text{fcn}} \in \mathbb{R}^{D_{\text{fcn}}}$ are learnable parameters of the FC layer, $W_{\text{cls}} \in \mathbb{R}^{D_{\text{cls}} \times D_{\text{fcn}}}$ and $b_{\text{cls}} \in \mathbb{R}^{D_{\text{cls}}}$ are learnable parameters of the classifier, $D_{\text{fcn}} = 256$ is the FC output dimension, and $D_{\text{cls}} = 8$ is the output dimension of the classifier.

Model training is performed using the standard cross-entropy loss, which measures the discrepancy between predicted class probabilities and ground-truth labels $g_n \in \{0, 1, \ldots, D_{\text{cls}} - 1\}$:

$$J_{\text{CE}} = -\frac{1}{B_{\text{in}}}\sum_{n=1}^{B_{\text{in}}} \log \widehat{p}_{(n,g_n)} \tag{40}$$

Minimizing J_{CE} encourages the model to assign high probability to the correct class for each input sample.

IV. EXPERIMENTS AND EVALUATION

To validate the proposed method, this section presents the experimental design and evaluation methodology. Comparative analyses conducted under diverse conditions are used to assess the robustness and efficiency of the proposed approach across multiple performance scenarios.

A. EXPERIMENTAL SETUP

1) TESTBED AND PROCESSING SYSTEMS

Following prior studies [45], [46], the experimental setup comprised two shafts, one located at the driven end and the other at the non-driven end, connected through a gearbox with a damping ratio of 1.52:1. Both shafts were mounted on a dedicated testbed and supported by cylindrical roller bearings (FAG NJ206-E-TVP2), whose specifications are summarized in Table 2. As illustrated in Fig. 6, the non-driven shaft was coupled to an adjustable blade via a pulley to enable controlled load variation. The rotational speed of the three-phase induction motor was regulated using a variable-speed controller. To reduce shock transmission and ensure stable operation, the motor and driven shaft were connected through a flexible coupling.

TABLE 2. Detailed bearing specification for the experimental testbed.

Pitch diameter	46.5 mm
Rolling element diameter	9.0 mm
Number of rolling elements	13
Contact angle	0°
Inner defect frequency	7.7581 × rotational speed
Outer defect frequency	5.2419 × rotational speed
Roller defect frequency	2.4866 × 2 × rotational speed

A wideband AE sensor was mounted on the non-drive-end bearing to capture high-frequency signal responses. The AE signals were acquired through a PCI-2 acquisition board configured with a decimated sampling rate of 250 kHz (the signals were downsampled by a factor of 4 using the *decimate()* function in MATLAB, which applies a built-in anti-aliasing filter). The PAC WSα sensor employed in this study provides an effective operating frequency range of 100-900 kHz, exhibits directional variation within ±1.5 dB, offers a peak sensitivity of -62 dB, and has a resonant frequency centered at 650 kHz.

All experiments were conducted using a high-performance workstation equipped with an AMD Ryzen 9 5950X processor (16 cores), 128 GB of system memory, and an NVIDIA GeForce RTX 3090 graphics processing unit (GPU). The experimental workflow integrated MATLAB and PyTorch at different stages. MATLAB was used for raw signal preprocessing and spectrogram generation, while PyTorch was employed for model implementation and training.

2) DATASET DESCRIPTION

To simulate bearing defects, cracks of varying dimensions were introduced using precision diamond cutting at different positions, resulting in distinct fault types as shown in Fig. 7: inner fault (BCI), outer fault (BCO), roller fault (BCR), inner-outer fault (BCIO), inner-roller fault (BCIR), outer-roller fault (BCOR), and inner-outer-roller fault (BCIOR). The constructed dataset included three crack sizes (3 mm, 6 mm, 12 mm) distributed across seven single-fault and compound-fault scenarios, and one healthy condition (BNC). Each case was evaluated at six rotational speeds (250 RPM, 300 RPM, 350 RPM, 400 RPM, 450 RPM, and 500 RPM) to ensure sufficient operational variability.

The recorded AE signals were segmented into 0.05-s intervals to generate individual samples (Fig. 8). It is important to note that segmentation was performed independently for each AE signal, and no segment overlaps with segments from

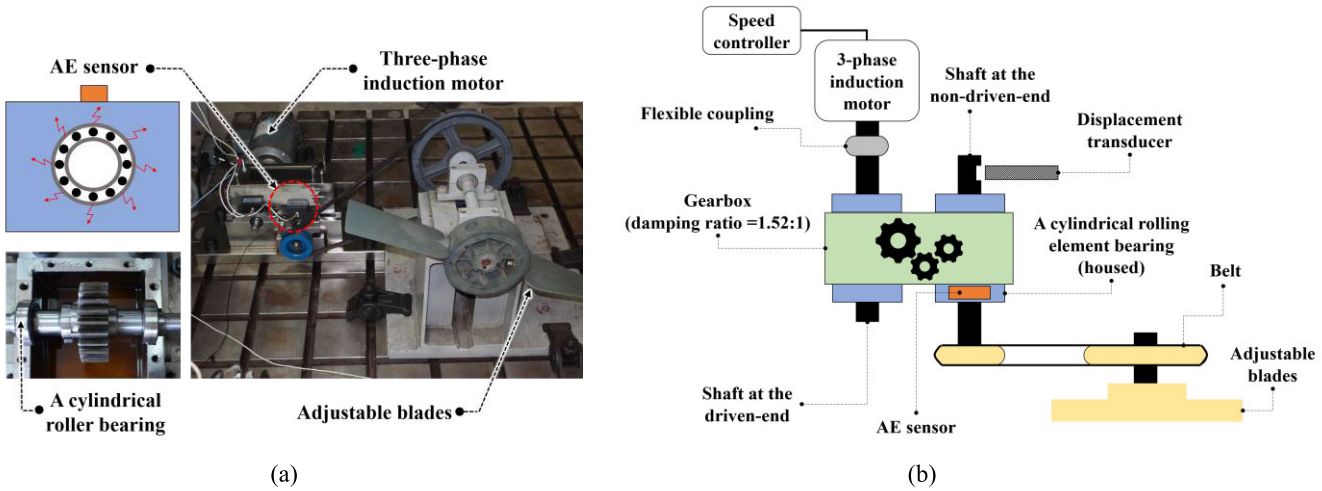


FIGURE 6. Experimental testbed and schematic representation. (a) Physical testbed configuration enabling precise monitoring and analysis of bearing faults under controlled mechanical and operational conditions. (b) Diagram of testbed, highlighting the motor-gearbox assembly and measurement instruments (AE sensor and displacement transducer) that facilitate signal acquisition across diverse operating scenarios.



FIGURE 7. Bearing fault types. (a) Outer fault (BCO). (b) Inner fault (BCI). (c) Roller fault (BCR). (d) Inner-outer fault (BCIO). (e) Outer-roller fault (BCOR). (f) Inner-roller fault (BCIR). (g) Inner-outer-roller fault (BCIOR).

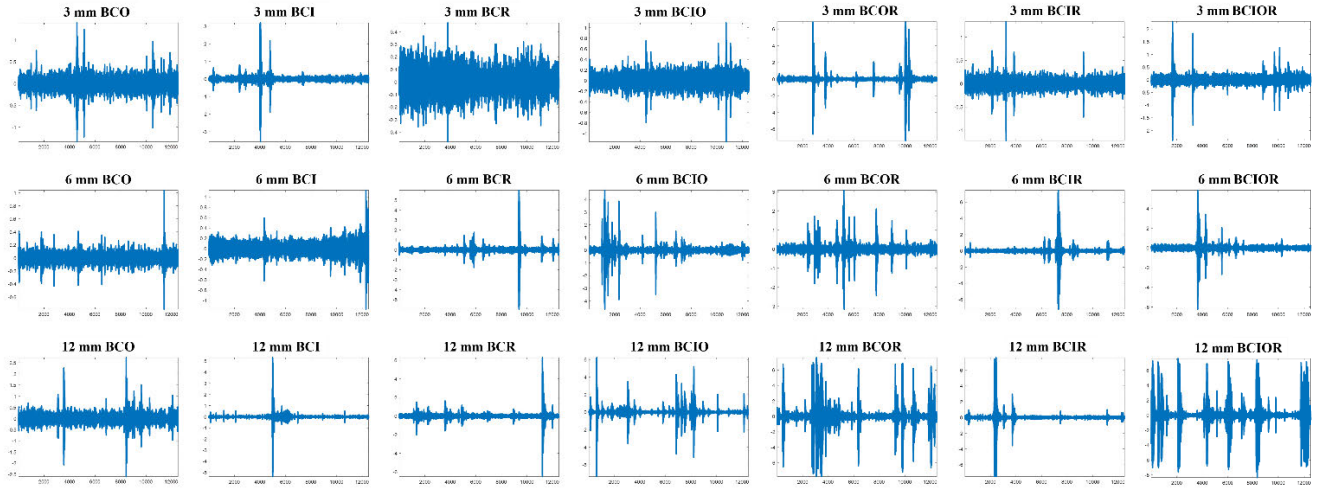


FIGURE 8. Segmented AE signals for various crack dimensions (3 mm, 6 mm, and 12 mm) and different bearing fault types at a rotational speed of 500 RPM.

different signal recordings. Each segment underwent signal processing to generate a corresponding spectrogram (Fig. 9). All transformations were performed independently for each segment without any global normalization or dataset-level statistical operations, ensuring that no information was shared across samples. For model development, the dataset was partitioned to support robust training, validation, and testing under diverse operating conditions. The training set comprises 13,200 samples, collected at rotational speeds of 300, 400, and 500 RPM. The validation set included 3,300 samples from the same fault categories but recorded at intermediate speeds of 250, 350, and 450 RPM to support unbiased model tuning. The testing set consisted of 6,600 samples, also obtained at 250, 350, and 450 RPM, and served

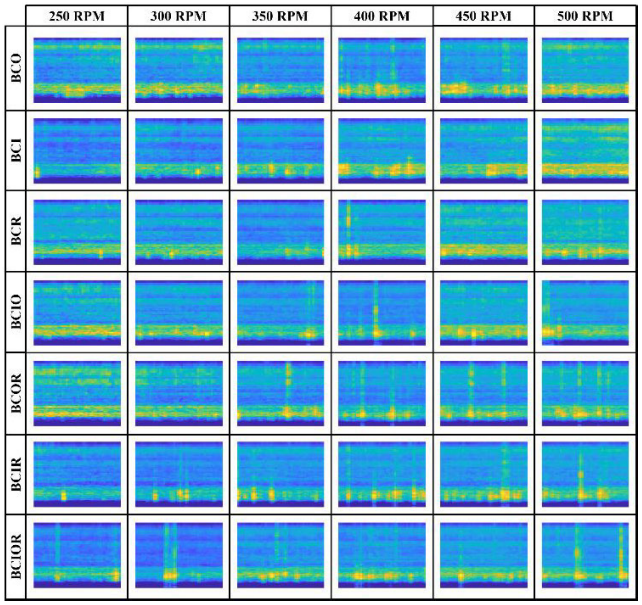


FIGURE 9. Representative log-Mel spectrograms of different bearing fault conditions with a 6 mm crack size, recorded at six rotational speeds (250 RPM, 300 RPM, 350 RPM, 400 RPM, 450 RPM, and 500 RPM).

as an independent benchmark for evaluating generalization performance.

3) TRAINING SETTINGS

Model training employed the AdamW optimizer, with hyper-parameters set to $\beta_1 = 0.9$, $\beta_2 = 0.999$, a weight decay of 1×10^{-4}, $\varepsilon = 1 \times 10^{-8}$. The learning rate was initialized at 5×10^{-4} controlled using a learning-rate scheduling strategy. Specifically, a warm-up phase of 1000 steps was applied, followed by polynomial decay with a power factor of 0.2 until reaching a minimum learning rate of 1×10^{-7}. Training was performed with a batch size of 256 for 30 epochs, providing sufficient iterations for learning discriminative features while mitigating overfitting.

B. EXPERIMENTAL EVALUATION

1) EVALUATION METRICS

To comprehensively evaluate classification performance across all fault classes, multiple standard metrics were employed. Overall accuracy was used to measure the proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{41}$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. The true positive rate (TPR), also referred to as recall, quantifies the proportion of correctly identified samples among all samples belonging to a given class:

$$TPR = \frac{TP}{TP + FN} \tag{42}$$

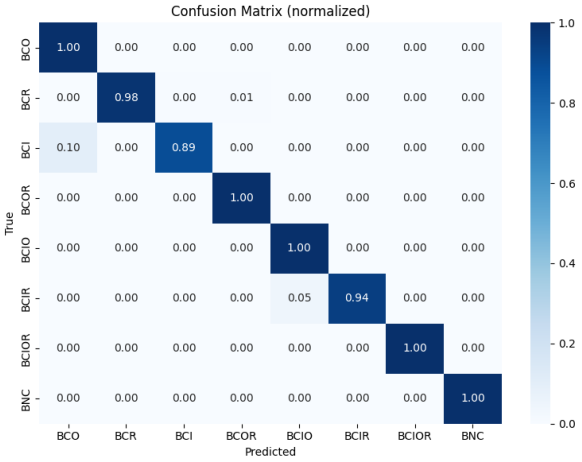


FIGURE 10. Normalized confusion matrix for the proposed method on the testing set.

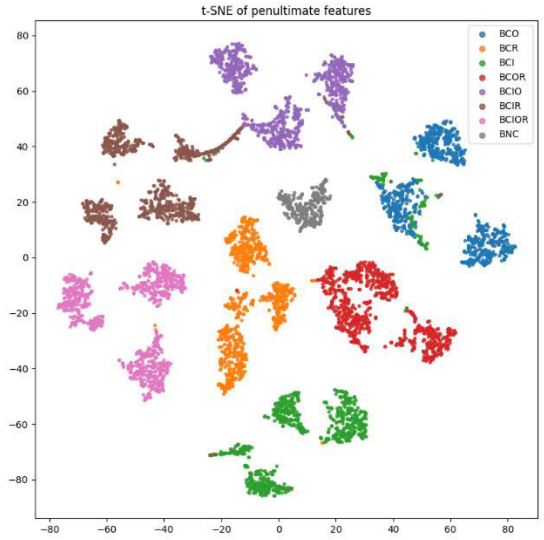


FIGURE 11. Feature space visualization using t-SNE for the proposed method on the testing set.

Precision also serves as a robust metric for evaluating the reliability of model performance:

$$Precision = \frac{TP}{TP + FP} \tag{43}$$

The $F1$ score, defined as the harmonic mean of $Precision$ and TPR, provides a balanced assessment of classification performance:

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{44}$$

The area under the curve (AUC) is a standard metric for multi-class classification and is derived from the receiver operating characteristic (ROC) curve, which plots the TPR against the false positive rate (FPR) across varying decision thresholds:

$$FPR = \frac{FP}{FP + TN} \tag{45}$$

$$AUC = \int_0^1 TPR(FPR)d(FPR) \tag{46}$$

This integral captures the overall ability of the classifier to discriminate between classes. A higher AUC value indicates better classification performance, with a value of 1.0 representing perfect discrimination and 0.5 corresponding to random guessing.

2) CLASSIFICATION PERFORMANCE

To assess the effectiveness of the proposed method, we analyze its classification behavior using the normalized confusion matrix and a t-SNE visualization of the penultimate features.

The normalized confusion matrix (Fig. 10) shows per-class recall. Most classes were classified almost perfectly, with TPRs exceeding 0.94 for all but one fault class (BCI). The healthy condition (BNC) was never confused with any fault class, which is particularly desirable in fault diagnosis applications.

To further examine the learned feature structure, t-SNE was applied to the penultimate features and visualized in 2-D (Fig. 11). Each class formed one or compact clusters that were clearly separated from those of other classes. This clustering behavior indicates that the learned representations were highly discriminative prior to the final linear classification layer.

The proposed architecture achieved an average testing accuracy of 97.41% and near-perfect recognition across most classes, including the BNC condition. The penultimate features formed class-specific manifolds with meaningful internal structure, demonstrating that the model learned a robust and interpretable representation of the data.

3) ABLATION STUDY

To rigorously evaluate the contribution of individual components within the proposed architecture, a series of ablation experiments was conducted. Key modules were systematically isolated to assess their impact on overall performance. Specifically, the M-SSSM was replaced with a vision transformer (ViT) to evaluate alternative global feature extraction strategies. To examine the role of the multiscale CNN, the M-SSSM was removed entirely. The proposed model was also benchmarked against lightweight CNN architectures commonly used in embedded systems. Finally, the standalone performance of the M-SSSM was evaluated to assess its independent efficacy.

As shown in Fig. 12, ResNet18 exhibited the fastest convergence, with training loss decreasing sharply within the first ∼150–200 steps and training accuracy approaching 1.0 early in training. The proposed multiscale CNN, multiscale CNN–ViT, and the proposed hybrid model converged at comparable rates, whereas MobileNetV2 and the M-SSSM converged more slowly. In terms of validation metrics, the proposed multiscale CNN and hybrid model outperformed MobileNetV2 and M-SSSM while demonstrating stronger generalization capability than ResNet18.

To confirm consistency, all experiments were repeated multiple times. Table 3 compares the validation accuracy and efficiency of the evaluated architectures. The proposed hybrid model with adaptive fusion achieved the highest mean accuracy (97.34% $\pm$ 0.44) and the highest maximum validation accuracy (98.09%), outperforming all other models at a reasonable computational cost. In contrast, MobileNetV2 and the M-SSSM yielded substantially lower mean accuracies, while ResNet18 achieved a comparable mean accuracy to the proposed multiscale CNN but with significantly higher complexity. From an efficiency perspective, the proposed models provided a favorable balance between accuracy and computational demand. The multiscale CNN achieved high accuracy while reducing parameter count by 50% relative to MobileNetV2 and 96.9% relative to ResNet18, offering a compact architecture that outperforms established lightweight baselines. Notably, ResNet18 was the most computationally demanding model and provided no accuracy advantage over the tailored multiscale design. These results highlight the importance of multiscale representations, which deliver superior performance with lower computational overhead. This advantage is particularly evident in the analysis of AE signals, where repetitive degradation patterns manifest across varying characteristic widths. By effectively capturing these multi-resolution features, the proposed approach demonstrates superior adaptability compared to baseline CNNs, which are primarily optimized for general-purpose tasks and lack the specialized capacity to model such scale-dependent signal variations. Moreover, adaptive fusion consistently outperformed static fusion in both mean and maximum accuracy with nearly identical computing cost, highlighting the value of data-dependent feature weighting. The rationale is that adaptive fusion dynamically modulates the contribution of each branch according to the input characteristics, selectively emphasizing the most informative representation while suppressing less reliable features. The proposed hybrid model was also markedly more efficient than the multiscale CNN + ViT, despite the growing popularity of CNN–transformer hybrids. This advantage arises from inherent design differences, whereas ViT depends on computationally intensive self-attention mechanisms and requires the input image to be partitioned into numerous smaller patches for processing, the M-SSSM architecture operates directly on the image without such preprocessing overhead.

Increasing the number of stacked layers in M-SSSM from 16 to 24 and 32 generally leads to higher GPU power consumption (Fig. 13). The 16-layer configuration operates mostly in the range of approximately 190–220 W, with several sudden drops near 100 W. In contrast, the 24-layer and especially the 32-layer models sustain higher average power levels, frequently exceeding 220 W and occasionally reaching peaks near 275–280 W. This trend indicates that deeper M-SSSM branches impose heavier computational workloads, resulting in higher GPU utilization and greater energy demand. The 32-layer model exhibits the highest and most sustained power draw, suggesting near-continuous execution pressure on the GPU resources. The streaming multiprocessor (SM) clock speed results provide further insight into

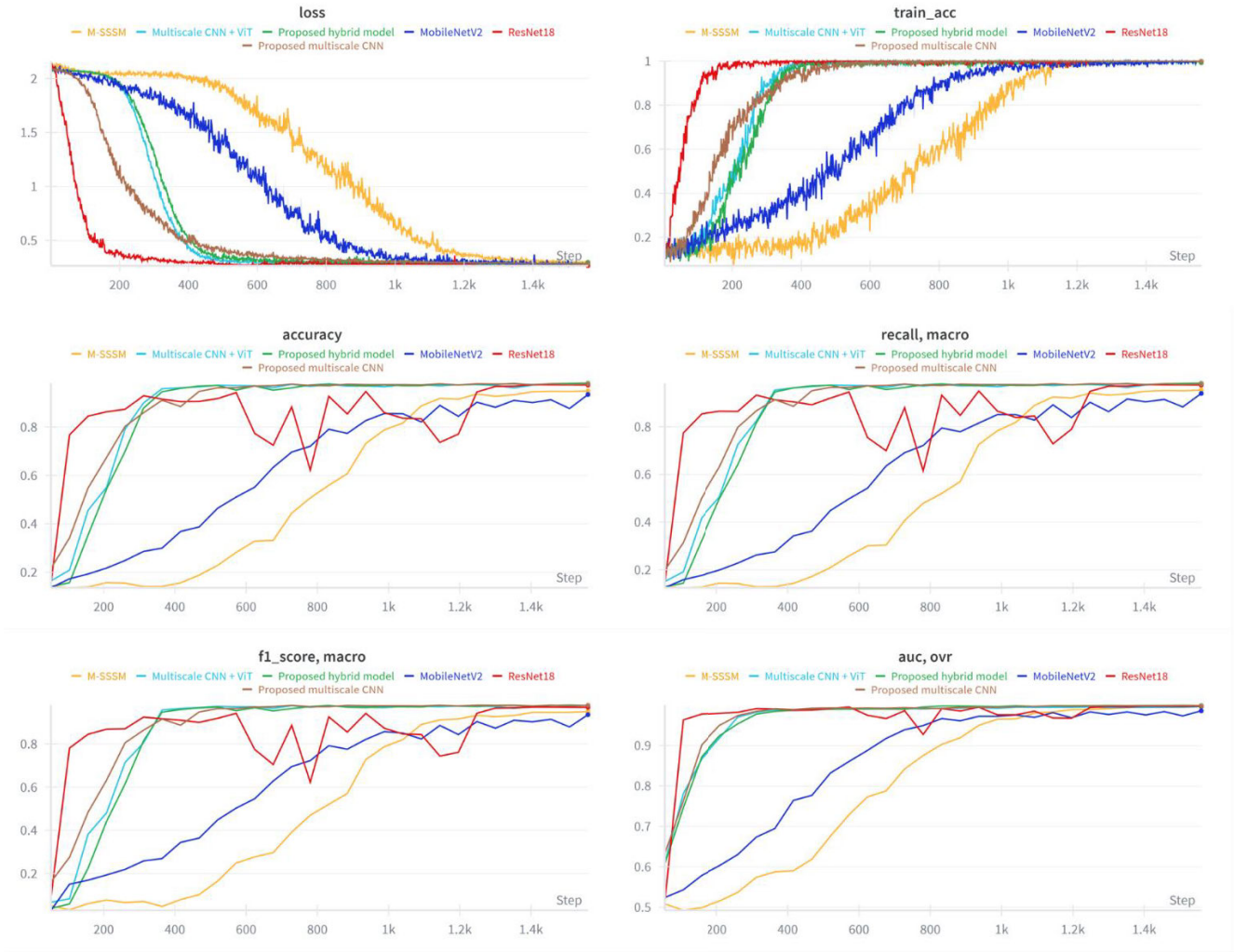


FIGURE 12. Ablation experiments are carried out to investigate the impact of different architectural elements on overall performance. The comparative settings include: (1) replacing the M-SSSM with a ViT, (2) removing the M-SSSM, (3) replacing the proposed model with baseline CNN architectures commonly used in embedded systems, and (4) removing the multiscale CNN in the hybrid model. The graphs show the comparative results of various evaluation metrics throughout the training and validation processes.

TABLE 3. Results of ablation study on diagnostic accuracy and computational efficiency.

Model	Mean validation accuracy (%)	Max validation accuracy (%)	Total parameters	Total MAC (GB)
MobileNetV2	92.073571 ± 1.106125	93.394	697,640	0.12
ResNet18	96.748857 ± 0.384353	97.273	11,171,784	4.97
M-SSSM	93.658143 ± 0.845404	94.879	339,976	0.05
Proposed multiscale CNN	96.748857 ± 0.514149	97.727	346,699	1.08
Proposed hybrid model (static fusion)	97.129714 ± 0.210298	97.333	750,411	1.14
Proposed hybrid model (adaptive fusion)	97.337714 ± 0.442666	98.091	849,099	1.15
Proposed multiscale CNN + ViT	96.986857 ± 0.48153	97.727	2,117,043	1.20

computational efficiency (Fig. 14). The 32-layer model maintains a highly stable clock frequency close to the maximum operating range (around 1900 MHz) throughout the experiment, with only minor fluctuations. This implies that the workload is sufficiently dense and regular to keep the GPU fully engaged, allowing boost clocks to remain stable. The 24-layer model also exhibits similar behavior but with occasional moderate drops, indicating transient underutilization

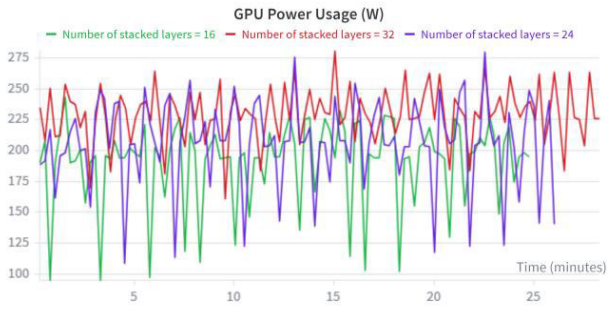


FIGURE 13. Comparison of GPU power usage of different stacked layers in the M-SSSM.

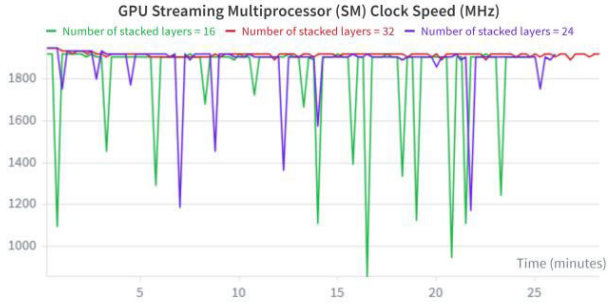


FIGURE 14. Comparison of the SM clock speed of different stacked layers in the M-SSSM.

TABLE 4. Ablation study on different numbers of stacked residual blocks in the proposed model.

Number of stacked blocks	Mean validation accuracy (%)	Total parameters	Total MAC (GB)
32	97.79	960,011	1.16
24	97.34	849,099	1.15
16	97.24	740,363	1.13

or memory stalls. In comparison, the 16-layer model experiences frequent and severe clock throttling events, with several drops below 1500 MHz and one instance below 900 MHz. Such instability suggests intermittent idle periods, reduced parallel workload saturation, or kernel launch inefficiencies when the M-SSSM branch is too shallow.

From an architectural perspective, these results suggest that increasing the number of stacked residual blocks improves hardware occupancy and computational continuity. A shallow M-SSSM branch may not generate a sufficient sequential modelling workload to fully exploit the GPU's parallel resources, causing clock oscillations and lower average power draw. Conversely, deeper configurations provide a more sustained workload, enabling more consistent SM utilization. However, this improvement comes at the cost of increased energy consumption. While the 32-layer configuration appears computationally efficient from a hardware utilization standpoint, it also consumes the most power. Therefore, selecting the optimal depth should consider both predictive accuracy and hardware efficiency. Table 4 shows that the accuracy gains from 32 layers are marginal relative to those of 24 layers. Accordingly, the 24-layer configuration represents a more favorable trade-off between performance and energy consumption.

The efficiency of different time–frequency analysis methods was further evaluated by measuring the processing time required for a single segmented AE signal. The continuous wavelet transform (CWT) [47] is implemented in MATLAB, which automatically selects the number of scales according to the signal length and sampling frequency. While this provides a practical and commonly used configuration, it should be noted that the computational cost of CWT is sensitive to the number of scales. Therefore, the reported runtime reflects this default setting rather than an exhaustive optimization of parameters. The HHT [48] is also performed in MATLAB with adaptive EMD, in that the number of intrinsic mode functions (IMFs) and the stopping criteria are determined based on the signal characteristics using the default MATLAB settings. The Wigner–Ville distribution (WVD) [49] is implemented using the default function in MATLAB, whereas the S-transform [50] is based on an open-source project [51]. Fig. 15 summarizes the comparative runtimes of five time–frequency techniques. The WVD and S-transform exhibited the highest computational cost. The HHT required moderate processing time due to the overhead associated with EMD and instantaneous frequency estimation. The CWT was faster when appropriately parameterized. This comparison shows that the STFT exhibits a lower computational cost compared to the other techniques under the current implementation settings. In contrast, the computational cost of CWT and HHT is highly dependent on parameter configurations, such as the number of scales and IMFs, leading to a trade-off between time-frequency resolution and efficiency.

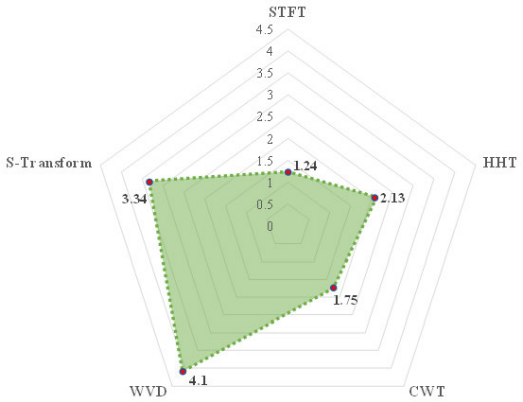


FIGURE 15. Comparison of various time-frequency analysis techniques applied to one segmented AE signal. The processing time of each method is scaled by using logarithmic normalization. The results show that STFT is the most efficient approach under the settings of this study, while S-Transform and WVD require huge computational complexity.

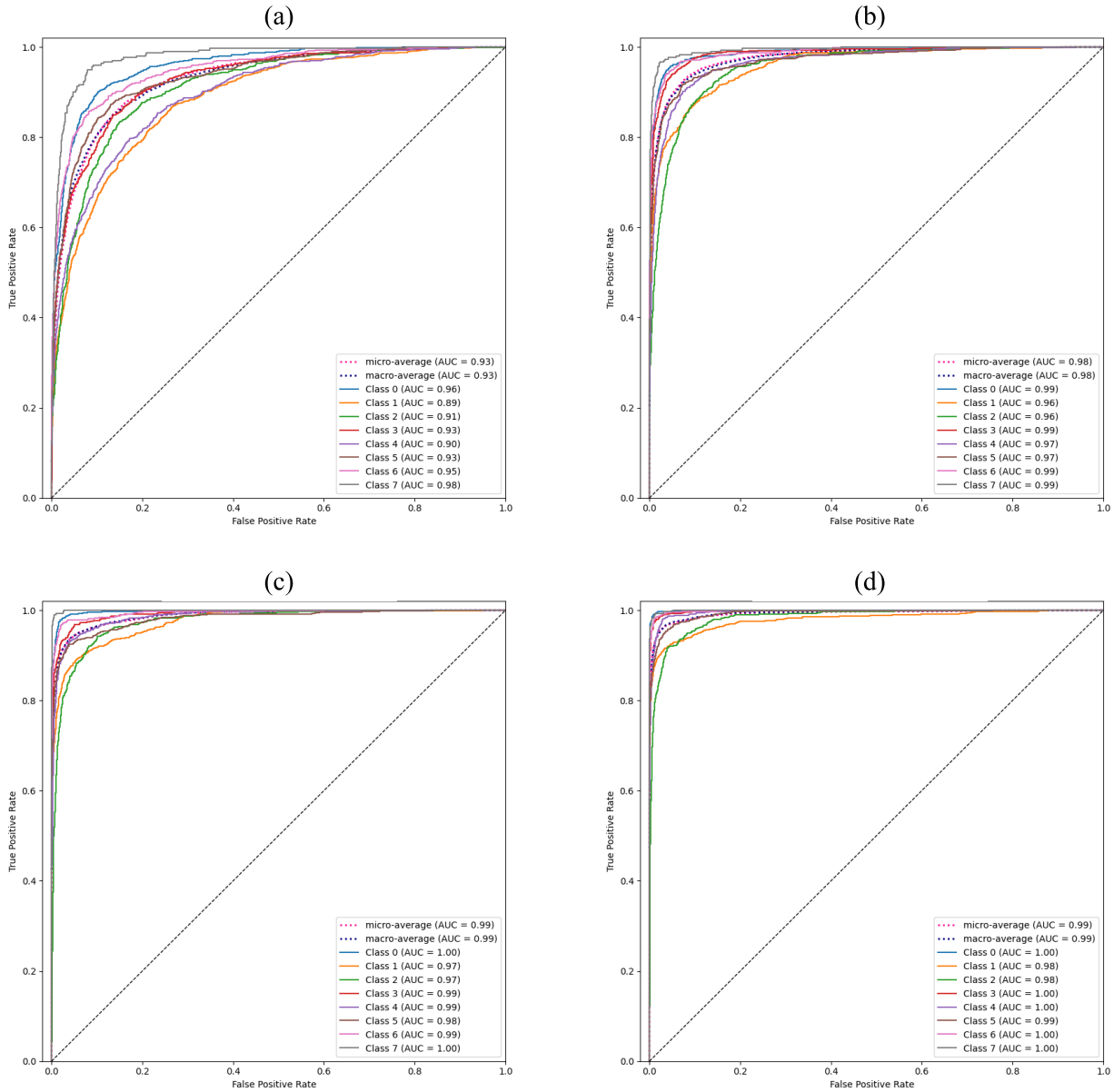


FIGURE 16. ROC curves illustrating classification performance of the proposed hybrid architecture under varying SNR conditions: (a) 0 dB, (b) 5 dB, (c) 10 dB, (d) 15 dB. Each curve highlights the impact of increasing SNR on detection accuracy and robustness. The proposed model achieves reliable discrimination between positive and negative instances at SNR levels of 10 dB and 15 dB.

4) PERFORMANCE IN NOISY CONDITIONS

To further assess robustness, the proposed bearing fault diagnosis method was evaluated under noisy conditions representative of industrial environments. The entire dataset was artificially contaminated with additive white Gaussian noise at signal-to-noise ratios (SNRs) of 0, 5, 10, and 15 dB. The experimental design enabled systematic evaluation of model resilience under increasing signal distortion. The SNR is defined as:

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \left(\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}} \right) \tag{47}$$

where P_{signal} denotes the power of the original signal and P_{noise} is the power of the added noise. Fig. 16 presents ROC curves under progressively cleaner signal conditions. As noise decreases, discrimination improves across all classes. At 0 dB (Fig. 16a), performance was modest, with notable inter-class variability and several classes exhibiting AUC values in the range of 0.89–0.93, indicating that severe noise obscured class-specific features and reduced sensitivity at low FPRs. At 5 dB (Fig. 16b), average AUC values increased to 0.98, with most classes achieving values between 0.96 and 0.99, demonstrating that moderate noise reduction substantially improved separability. At 10 dB (Fig. 16c),

performance approached saturation (micro-/macro-AUC = 0.99/0.99), with two classes reaching perfect discrimination and the remaining classes tightly clustered between 0.97 and 0.99. At 15 dB (Fig. 16d), near-ceiling performance persisted, indicating minimal residual impact of noise. These results indicate that noise mitigation was critical at very low SNRs, while at typical operating SNRs above 10 dB, the proposed model consistently delivered robust classification performance.

We extend our framework to incorporate composite noise, which more accurately reflects the complexity of disturbances encountered in industrial environments. Unlike isolated noise models, composite noise captures the simultaneous presence of multiple stochastic processes, thereby offering a more representative approximation of real-world conditions. Specifically, the composite noise considered in this study comprises Gaussian, uniform, and impulse noise, each corresponding to distinct phenomena observed in manufacturing and signal acquisition systems. Gaussian noise is employed to represent the ubiquitous thermal fluctuations and electronic interference that arise in sensors. Uniform noise is introduced to simulate quantization effects inherent in digital signal acquisition. Impulse noise captures transient, high-amplitude disturbances such as mechanical shocks, electrical spikes, and intermittent interference in industrial environments. The composite noise used in this study is mathematically represented as:

$$\begin{aligned} r^{\text{noise}} = r^{\text{raw}} &+ \mathbb{N}\left(0, \frac{P_{\text{signal}}}{10^{\frac{\text{SNR}_{\text{dB}}}{10}}}\right) \\ &+ \mathbb{U}\left(-\sqrt{3 \cdot P_{\text{noise}}}, \sqrt{3 \cdot P_{\text{noise}}}\right) \\ &+ \text{Bernoulli}\,(p) \cdot \mathbb{N}\left(0, \frac{P_{\text{noise}}}{p}\right) \end{aligned} \tag{48}$$

where r^{raw} is the raw AE signals, r^{noise} is the contaminated AE signals, $\mathbb{N}$ represents Gaussian noise, $\mathbb{U}$ denotes uniform distribution, and p is the Bernoulli success probability.

Uniform noise is generated as a zero-mean random process with an amplitude range $[-\sqrt{3 \cdot P_{\text{noise}}}, \sqrt{3 \cdot P_{\text{noise}}}]$, ensuring that the expected noise power equals P_{noise}. Impulse noise is generated using a Bernoulli-Gaussian model. A Bernoulli mask with $p = 0.005$ selects corrupted samples, and the selected samples are replaced by Gaussian noise scaled by $\sqrt{P_{\text{noise}}/p}$, ensuring constant expected noise power.

The proposed method demonstrates consistent robustness across increasing SNR levels, with classification accuracy improving monotonically for all noise types as the noise level decreases (Fig. 17). This indicates that the extracted representations become progressively more discriminative as signal corruption decreases. Impulse noise yields the highest accuracy at all tested SNR levels, achieving approximately 86.67%, 90.08%, 94.38%, and 95.55% at 0, 5, 10, and 15 dB, respectively. This suggests that the model is comparatively resilient to sparse impulsive disturbances. Because impulse noise affects only a small subset of samples, much of the original signal structure remains intact. In addition, transient AE fault signatures are inherently impulsive in nature, which may make the model less sensitive to sparse spike-like corruption than to dense stochastic noise. Fig. 17 also shows that Gaussian noise produces intermediate performance. Compared with impulse noise, Gaussian noise perturbs all samples continuously, which can blur weak fault-related patterns in both the time and time-frequency domains. However, the steady improvement with SNR indicates that the proposed framework retains good robustness to additive white Gaussian noise. Composite noise represents the most challenging condition, yielding the lowest accuracy across all SNR levels. This is expected because composite noise combines multiple degradation mechanisms simultaneously, causing both distributed contamination and occasional high-amplitude disturbances. Such mixed corruption more severely distorts diagnostic features and reduces class separability.

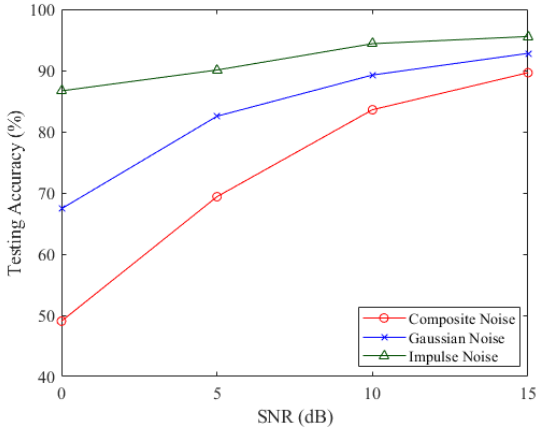


FIGURE 17. Testing accuracy under different noise conditions (composite, Gaussian, and impulse noise) across SNR levels ranging from 0 to 15 dB.

Overall, the results verify that the proposed method maintains strong diagnostic capability under harsh noisy environments, particularly under impulse and Gaussian noise, while composite noise remains the most demanding scenario. These findings also highlight the importance of evaluating bearing fault diagnosis systems under heterogeneous noise conditions, rather than relying solely on Gaussian assumptions.

5) EDGE DEPLOYMENT

To evaluate real-world applicability, the proposed model was deployed on an embedded platform, specifically the NVIDIA Jetson Nano, which features a quad-core ARM Cortex-A57 processor, 4 GB of LPDDR4 memory, and a 128-core NVIDIA Maxwell GPU. During inference, performance was assessed by recording maximum power consumption and peak GPU shared-memory utilization. Fig. 18 compares resource usage across input resolutions of 128 × 128, 64 × 64, and 32 × 32 for ResNet18, multiscale CNN + ViT, and the proposed model.

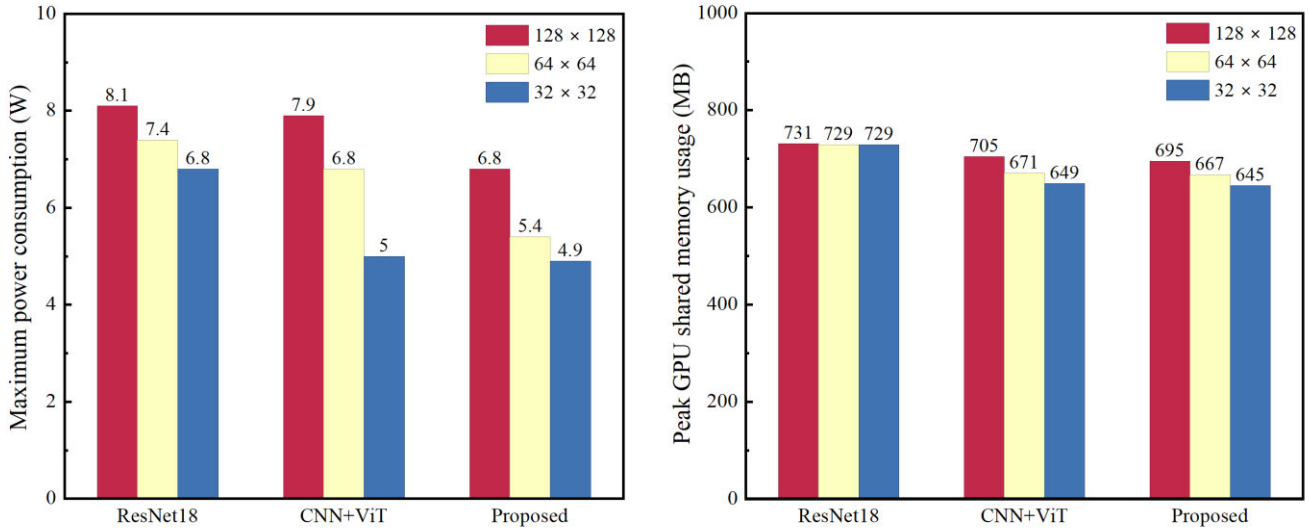


FIGURE 18. Performance comparison on edge device across varying input resolutions. The proposed model consistently demonstrates superior efficiency, achieving the lowest power consumption and memory utilization when compared with both the recent hybrid CNN+ViT approach and the widely adopted baseline ResNet18.

The proposed model consistently exhibited the lowest power consumption across all resolutions, decreasing from 6.8 W at 128 × 128 to 5.4 W at 64 × 64 and 4.9 W at 32 × 32. The largest gains occurred at the highest resolution, where peak power was reduced by 1.3 W (∼16%) relative to ResNet18 and by 1.1 W (∼14%) relative to multiscale CNN + ViT. At 32 × 32 resolution, the proposed model and multiscale CNN + ViT showed comparable power consumption, reflecting the relatively low overhead of ViT at small input sizes. These results show that the M-SSSM is more efficient compared to ViT when processing larger input sizes or longer sequences, thanks to the selective scan mechanism. In contrast, ResNet18 maintained higher power consumption, indicating poorer scalability at low resolutions.

Peak GPU shared-memory utilization followed a similar trend. The proposed model consistently used the least memory across all resolutions, with the largest relative savings observed at 128 × 128 (−4.9%) compared with ResNet18. These advantages persisted as resolution decreased. Overall, the results demonstrated that the proposed architecture offers the most favorable resource profile, achieving minimal power draw and memory usage across all evaluated input resolutions. These characteristics make it particularly well-suited for edge deployments subject to strict energy and memory constraints.

V. CONCLUSION

In this study, we proposed a novel lightweight heterogeneous architecture for efficient bearing-fault diagnosis. By integrating the proposed multiscale CNN with a customized M-SSSM, the framework outperforms recent CNN–ViT hybrid approaches and conventional CNN baselines in both diagnostic accuracy and computational efficiency under practical operating conditions. In addition, the results demonstrate that the STFT-based time–frequency analysis employed in this work is well-suited for resource-constrained environments.

Despite these contributions, several limitations warrant further investigation. First, the current evaluation is based on a relatively constrained dataset, which may not fully capture the operating conditions encountered in large-scale industrial environments. Expanding the dataset in both size and diversity will be essential to validate the generalizability of the proposed approach. Second, the experiments were conducted on a single device configuration, leaving open questions regarding device generalization, that is, whether the model maintains comparable performance across different sensor types, acquisition hardware, and computational platforms.

Moreover, the signal processing and inference stages are currently implemented as separate modules rather than within a unified, end-to-end pipeline, which may limit optimization opportunities. The evaluation also does not include explicit measurements of inference latency, a critical consideration for real-time industrial applications. Addressing these limitations, through larger and more diverse datasets, systematic cross-device and cross-platform validation, and integrated pipeline optimization, will further enhance the robustness and practical applicability of the proposed approach in large-scale industrial deployments.

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